

Strict Positive Square Energy for Two-Connected Noncycle Graphs

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Abstract

For a graph G , let $s^+(G)$ be the sum of the squares of its positive adjacency eigenvalues. Akbari, Hu, and Liu proved that $s^+(G) \geq |V(G)|$ for every two-connected graph that is not a cycle, leaving equality possible in the part of their argument involving triangles. We rule out that possibility. The key observation is that if P is the positive spectral part of the adjacency matrix, then every triangle contains a vertex whose incident squared mass in P is strictly greater than 2. This follows by combining the equality case of a six-term Cauchy–Schwarz estimate with the orthogonality of the positive and negative spectral parts. Consequently $s^+(G) > |V(G)|$ for every finite simple two-connected noncycle graph. Thus the equality class in this domain is empty.

1 Context and statement

All graphs in this note are finite, simple, and undirected. Let $A = A(G)$ be the adjacency matrix of a graph G and let its eigenvalues, counted with multiplicity, be $\lambda_1, \dots, \lambda_n$. Define

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2.$$

The connected-graph square-energy theorem of Liu, Tang, and Zhang states that, whenever G is connected,

$$\min\{s^+(G), s^-(G)\} \geq n - 1$$

[2]. Akbari, Hu, and Liu used this bound together with a vertex-deletion inequality to prove $s^+(G) \geq n$ for every two-connected noncycle G ; they obtained strictness when some maximum-degree vertex lies in no triangle, and hence in particular for triangle-free graphs [1, Theorem 3.4].

We close exactly the remaining equality case.

Theorem 1. *Let G be a finite simple two-connected graph of order n . If G is not isomorphic to the cycle C_n , then*

$$s^+(G) > n.$$

Thus the requested structural classification has a particularly simple form: there are no equality graphs in the stated domain. The contribution is the strict local triangle estimate in theorem 2; the other ingredients are the cited connected-graph bound and the triangle-free clause of Akbari–Hu–Liu. This is a comparison with those inspected sources, not a claim of global priority.

2 Spectral parts and deletion

For a real symmetric matrix A , write

$$A = P - N, \quad P = A_+ \succeq 0, \quad N = A_- \succeq 0, \quad PN = 0,$$

where positive and negative eigenvalues are retained in P and N with their absolute values. In particular, $s^+(G) = \text{tr}(P^2) = \|P\|_F^2$. For $v \in V(G)$ set

$$\mu_v(P) = P_{vv}^2 + 2 \sum_{u \neq v} P_{vu}^2.$$

We will use the deletion inequality

$$s^+(G) \geq s^+(G - v) + \mu_v(P), \quad (1)$$

which also appears as Lemma 2.4 of [1]. For completeness, it follows directly from the positive-part decomposition. Indeed, with $T = V(G) \setminus \{v\}$, the adjacency matrix $B = A[T]$ has a decomposition $B = P[T] - N[T]$ into two positive semidefinite matrices. If B_+ is its positive spectral part, then

$$\|B_+\|_F^2 = \text{tr}(BB_+) \leq \text{tr}(P[T]B_+) \leq \|P[T]\|_F \|B_+\|_F.$$

Hence $s^+(G - v) = \|B_+\|_F^2 \leq \|P[T]\|_F^2$, while $\|P[T]\|_F^2 = s^+(G) - \mu_v(P)$, proving (1).

3 A strict triangle estimate

Lemma 2 (Strict triangle lemma). *If $S \subseteq V(G)$ is the vertex set of a triangle, then some $v \in S$ satisfies $\mu_v(P) > 2$. No connectedness assumption is needed.*

Proof. Put $X = P[S]$. For $v \in S$, define the part of the local mass supported on the triangle by

$$\tau_v = P_{vv}^2 + 2 \sum_{u \in S \setminus \{v\}} P_{vu}^2.$$

Since $P - A = N \succeq 0$, its principal submatrix on S is positive semidefinite. Because $G[S] = K_3$,

$$\mathbf{1}^\top X \mathbf{1} \geq \mathbf{1}^\top A[S] \mathbf{1} = 6. \quad (2)$$

Apply Cauchy–Schwarz to the six real numbers

$$P_{vv} \quad (v \in S), \quad 2P_{uv} \quad (\{u, v\} \in \binom{S}{2}).$$

Their sum is $\mathbf{1}^\top X \mathbf{1}$, and the sum of their squares is $\sum_{v \in S} \tau_v$. Therefore

$$36 \leq (\mathbf{1}^\top X \mathbf{1})^2 \leq 6 \sum_{v \in S} \tau_v. \quad (3)$$

Suppose for a contradiction that $\mu_v(P) \leq 2$ for every $v \in S$. Since $\tau_v \leq \mu_v(P)$, (3) forces equality throughout: $\tau_v = \mu_v(P) = 2$ for every $v \in S$. Equality in Cauchy–Schwarz, together with (2), forces all six displayed numbers to be 1. Hence

$$X = \begin{pmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{pmatrix}. \quad (4)$$

Moreover, for every $v \in S$,

$$0 = \mu_v(P) - \tau_v = 2 \sum_{u \notin S} P_{vu}^2,$$

so $P[S, V(G) \setminus S] = 0$. The eigenvalues of the matrix in (4) are $2, 1/2, 1/2$, so X is invertible. Taking the S -by- S block of $PN = 0$ now gives

$$0 = (PN)[S, S] = XN[S, S],$$

because the other block product vanishes. Thus $N[S, S] = 0$. On the other hand, $N = P - A$, and (4) together with $A_{vv} = 0$ gives $N_{vv} = 1$ for $v \in S$, a contradiction. $\square$

Proof of theorem 1. A two-connected noncycle has $n \geq 4$ and maximum degree at least 3. If G is triangle-free, the strict clause of [1, Theorem 3.4] gives $s^+(G) > n$.

Otherwise, choose a triangle S . By theorem 2, it contains a vertex v with $\mu_v(P) > 2$. Two-connectivity makes $G - v$ connected, so the Liu–Tang–Zhang bound gives $s^+(G - v) \geq (n - 1) - 1 = n - 2$. Using (1),

$$s^+(G) \geq s^+(G - v) + \mu_v(P) \geq n - 2 + \mu_v(P) > n.$$

The triangle-free and triangle cases exhaust the domain. $\square$

Corollary 3. *There is no finite simple two-connected noncycle graph G satisfying $s^+(G) = |V(G)|$.*

4 Finite corroboration and scope

The proof above is independent of computation. As a falsification check, one representative of each unlabeled two-connected graph of orders 4 through 8 was generated with **geng -C**. After excluding the one cycle at each order, the screen covered the following numbers of graphs.

Order	Two-connected	Noncycles	Equality candidates
4	3	2	0
5	10	9	0
6	56	55	0
7	468	467	0
8	7123	7122	0
Total	7660	7655	0

The spectral screen used 80-decimal-digit arithmetic. For all ten graphs whose observed gap $s^+(G) - |V(G)|$ was at most 1.01, integer characteristic polynomials and rational isolating intervals gave an exact strict certificate. The user-supplied bibliography describes the same two exact tools—integer characteristic polynomials and certified Sturm root counts—in a separate finite line-graph inertia verification by Zeng [3]; that citation is only a methodological comparison, not support for the theorem proved here. Finite enumeration in either setting is not evidence for an all-orders conclusion by itself.

The closest inspected result remains [1, Theorem 3.4], whose non-strict bound and restricted strict clause are strengthened here in the same two-connected noncycle domain. A bounded literature comparison did not locate the strict theorem in the inspected primary sources, but this is not an exhaustive search and no claim of priority is made.

Reproducibility. The accompanying files `enumerate_n4_n8.py` and `enumeration_n4_n8.json` contain the enumeration code, graph encodings, software versions, high-precision values, and rational root certificates. A rerun reproduced the JSON output byte for byte.

Computational and AI assistance. The finite check used nauty, NetworkX, mpmath, and SymPy. An OpenAI language model assisted with literature triage, proof auditing, code review, and LaTeX editing. The mathematical proof does not rely on model output or on floating-point computation; the author takes responsibility for all claims, citations, and exact certificates.

References

- [1] S. Akbari, Fu-Tao Hu, and Ya-Yang Liu. Positive and negative square energies of 2-connected graphs, 2026.
- [2] Yinchen Liu, Quanyu Tang, and Shengtong Zhang. The positive and negative square-energy conjecture, 2026.
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